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News and Views

New estimates of blood flow rates in the vertebral artery of euarchontans and their implications for encephalic blood flow scaling: A response to Seymour and Snelling (2018)

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1. Introduction

Variation in the size and structure of the brain is thought to have intense selective consequences via correlated variation in cognitive capacity and metabolic energy demands. While the relationships between brain size or structure, cognitive abilities, and brain metabolic costs are not straight forward (Willerman et al., 1991; Healy and Rowe, 2007; Karbowski, 2007, 2011; Shettleworth, 2009; Herculano-Houzel, 2011), it is usually thought that traits contributing to greater cognitive abilities (e.g., an absolutely larger brain) also incur higher metabolic cost (e.g., Aiello and Wheeler, 1995). This implies an evolutionary tradeoff in which adaptive benefits of increased cognition must be balanced against increased energy costs.

Boyer and Harrington (2018) added to the discussion about tradeoffs in brain evolution with a study which concluded that, compared with other animals, humans do not have exceptionally expensive brains relative to the basal metabolic rates of their bodies. In a recent technical comment, Seymour and Snelling (2018) suggested that the study of Boyer and Harrington (2018) overestimates the role of the vertebral artery in supplying the brain with blood and potentially mischaracterizes variation in brain metabolic demands as a result.

In this response, we address the methodological concerns of Seymour and Snelling (2018) and report results of re-analyzing the data of Boyer and Harrington (2018). This exercise shows that the ratio of internal carotid to vertebral blood flow rate tends to be higher than implied by the ratio of vessel canal cross-sectional dimensions reported by Boyer and Harrington (2018). However, we also found that the results of analyses supporting the major conclusions of Boyer and Harrington (2018) are robust to this offset, as well as to other methodological issues noted by Seymour and Snelling (2018). This response article also addresses a question that was outside of the scope of Boyer and Harrington's (2018) study. Specifically, an added benefit of modifying the methods of Boyer and Harrington (2018) according to Seymour and Snelling's (2018) critique is that we are able to present an estimate of the scaling relationship between encephalic blood flow rate and brain size. Our results indicate that blood flow rates to the brain scale with negative allometry for at least three mammalian groups of interest: (1) primates as a whole, (2) a more exclusive group representing only haplorhine primates (humans, apes, and other anthropoids, as well as tarsiers), and (3) the more inclusive grouping of primates, treeshrews and dermopterans (Euarchonta). In all of these groups, confidence limits on the scaling exponent exclude both isometry (1.0) and 'metabolic scaling' (Kleiber, 1947) (0.75). Instead, the mean exponent closely matches the exponent of 0.86 for scaling of brain metabolism to brain size previously documented for a more phylogenetically inclusive sample of eleven placental mammal species (Karbowski, 2007).

2. Background

Seymour and colleagues (2015, 2016, 2017, 2018) have published stimulating research estimating the interspecific scaling of blood flow rates to the brain. These studies explore the connection between blood flow rate scaling, brain metabolic expense, and cognitive complexity. Specifically, they have looked at variation in and scaling of blood flow rates across marsupials, primates (Seymour et al., 2015), and hominins (Seymour et al., 2016, 2017)

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using a new method to predict blood flow rates through the internal carotid artery (ICA) from the radius of the posterior carotid foramen based on the Poiseuille–Hagen equation. They consider the scaling of these predicted ICA flow rates to brain size as reflective of the scaling of blood flow rate to the forebrain. They found that haplorhine primates and diprotodont marsupials exhibit different scaling exponents and interpreted this pattern to reflect different levels of cognitive complexity. The shallow slope of marsupials was interpreted as reflecting reduced selection for increasing cognitive complexity in larger brains, whereas the higher slope of haplorhines (including fossil hominins) was said to reflect greater and/or increasing cognitive complexity over time. One of the most surprising implications of their work (Seymour et al., 2016, 2017) is that, during hominin evolution, the metabolic cost of the forebrain increased by six-fold even though forebrain size increased by only 3.5 times.

While Boyer and Harrington (2018) accepted Seymour's basic premise relating the osteological foramen for the ICA to blood flow and brain energy demands, they questioned certain aspects of Seymour et al.'s methods and interpretations. The equation Seymour et al. (2015, 2016, 2017) used for calculating flow rate from vessel size also requires information on vessel wall shear stress (τ). However, Seymour et al. (2015, 2016, 2017) had to estimate τ of the ICA based on species mean body mass (M_b) for the vast majority of taxa in their sample. In Seymour and colleagues' studies, the relationship between τ and M_b is derived from data on two taxa (*Rattus* and *Homo*) in which both ICA flow rates and vessel radii (both necessary to compute τ) are available in the literature. Because of the small amount of data on τ, Boyer and Harrington (2018) were concerned about the accuracy of the relationship between τ and M_b. Furthermore, Boyer and Harrington (2018) evaluated the effect of varying input parameter values for the τ equations on blood flow rate calculations. They were uncomfortable with the degree to which varying the parameter values in the τ equations affected blood flow rate scaling exponents in their sample.

To avoid perceived issues introduced by having to predict shear stress (τ) and blood flow rates, Boyer and Harrington (2018) attempted to develop an approach for predicting brain metabolism directly from the dimensions of the osteological canals (or foramina) for encephalic arteries. As a separate issue, Boyer and Harrington (2018) also questioned whether information on the ICA alone is sufficient for estimating the scaling relationship of blood flow rate to the forebrain since the vertebral artery can also supply the forebrain with blood, giving off the posterior cerebral vessels to the occipital lobe of the cerebrum, and anastomosing with the ICA in the circle of Willis.

Boyer and Harrington (2018) concluded that the vertebral artery contributes significantly to the blood requirements of the forebrain based on a significant partial correlation between transverse foramen cross-sectional area and forebrain mass, when accounting for variance in internal carotid canal area. This finding led them to question the validity of Seymour and colleagues' (2015, 2016) conclusion that different clades have different scaling relationships of blood flow to forebrain size as predicted from the ICA canal alone. Boyer and Harrington (2018) hypothesized that adding information on the vertebral artery would show more uniformity in scaling of blood flow to brain size. They tested this by summing cross-sectional areas of the ICA and vertebral artery canals (as a presumed correlate of total flow rate) and evaluating whether haplorhines differed from strepsirrhines in the scaling of brain size to encephalic arterial canal size. They found no differences in slope or intercept. Based on this finding, they speculated that the clades examined in Seymour et al.'s studies would become more uniform in their scaling exponents if data on the vertebral artery flow rate were added to that of the ICA flow rate, and that hominins would no longer show a scaling relationship of blood flow rate to brain size indicative of increasing cerebral perfusion with brain size.

In their recent technical comment on Boyer and Harrington (2018), Seymour and Snelling (2018) now point out some reasons that their flow rate calculations may be more accurate and reliable than Boyer and Harrington (2018) originally believed. Furthermore, Seymour and Snelling (2018) raise several concerns about the conclusions of Boyer and Harrington (2018). These concerns stem from Seymour and Snelling's (2018) observation that Boyer and Harrington's (2018) method does not 1) acknowledge that the scaling relationship of osteological canal area to brain size is necessarily disproportionate to that of blood flow rate to brain size, or 2) incorporate presumed differences in “canal filling” by the ICA and vertebral arteries. Specifically, Seymour and Snelling (2018) note that the vertebral artery is likely to fill the transverse foramen less completely than the ICA fills the carotid canal, citing literature on humans and monkeys (Epstein et al., 1970; Eckenhoff, 1971; Paullus et al., 1977; Vijaywargiya et al., 2017). They state that this leads to an over-estimation of the role of the vertebral artery in brain blood perfusion. Graciously, Seymour and Snelling (2018) acknowledge that Boyer and Harrington's (2018) data could be fruitfully used to re-visit the questions of vertebral artery contribution, scaling of blood flow rate to the brain, and brain metabolism.

We acknowledge the relevance of the above aspects of Seymour and Snelling's (2018) critique and have taken the opportunity reanalyze the data of Boyer and Harrington (2018) accordingly. Specifically, in this paper we aim to 1) use encephalic arterial canal diameter data from Boyer and Harrington (2018) to predict blood flow rate, and 2) re-run our analyses using blood flow rates in place of canal areas.

3. Methods

Our first step was to address the question of ‘canal filling’ (i.e., the degree to which a vessel's diameter matches the diameter of the osteological canal through which it passes). We searched the literature and did dissections of both perfused and fixed cadavers to evaluate how variable the vertebral artery is in the degree to which it fills the transverse foramen of the cervical vertebrae and whether it is consistently smaller relative to canal size than the ICA. Data on four species were available from the literature (Table 1A). We gathered new data for another five species (one individual each) through dissections on formalin-preserved cadaveric specimens. Four of these specimens are from the wet specimen collection of the Duke University Department of Evolutionary Anthropology. A specimen of *Rattus norvegicus* was purchased from Carolina Biological Supply Company. On each specimen, the vertebral arteries were exposed, photographed, and external diameters were measured from images. Lumen radius was then calculated as [(*external diameter* ÷ 2) ÷ 1.4]. Dividing by 1.4 adjusts for the expected vessel wall thickness as discussed in Seymour et al. (2015). None of the specimens was measured under physiological pressure, although two specimens (*Tupaia* sp. and *R. norvegicus*) were perfused with latex before we came into possession of them. Table 1A, B shows a summary of the data we obtained.

We found the average vertebral artery lumen radius for nine taxa to be 51% (±14%) of the bony transverse foramen radius (Table 1A), compared to 76% (±9%) for the internal carotid lumen radius (Table 1B), confirming Seymour and Snelling's (2018)

Table 1A
Measurements of vessel lumen and osteological foramen diameters for select species. Reliability refers to the method by which the lumen diameter was measured: 1 = from CT, x-ray, or synchrotron angiography images; 2 = a directly measured vessel on a newly dissected specimen in which the vessel was formalin preserved and perfused with latex; 3 = a directly measured vessel on a newly dissected specimen in which the vessel was formalin preserved, but not perfused. For 2 & 3, lumen radius was computed according to [(external diameter/2)/1.4] following Seymour et al. (2015). When more than one study is cited, measurements are averages of values reported in each source. Note that reliability rank 2 & 3 specimens values are likely minimum estimates of vessel lumen size due to postmortem shrinkage of vessels in formalin. Ratio = [lumen radius/foramen radius]. Sources: A—Tomasino et al. (2010), B—Findlay et al. (1988), C—Macdonald et al. (1991), D—Schambach et al. (2009), E—Jeppsson and Olin (1960), F—Boyer and Harrington (2018). Digital Object Identifiers (DOI) link to images measured to generate 'lumen radius' data.

Species & specimen image DOI	Vertebral artery			Transverse foramen		Ratio
	Reliability	Lumen radius (mm)	Sources	Radius (mm)	Sources	
Homo sapiens	1	1.69	A	2.7	A	0.63
Macaca fascicularis	1	0.51	B, C	1.1	F	0.46
Mus musculus	1	0.1	D	0.14	F, this study	0.71
Rattus norvegicus https://doi.org/10.17602/M2/M62069	2	0.22	this study	0.31	F	0.71
Oryctolagus cuniculus	1	0.34	E	0.97	F, this study	0.35
Sciurus carolinensis https://doi.org/10.17602/M2/M62070	3	0.29	this study	0.65	this study	0.45
Varecia variegata https://doi.org/10.17602/M2/M62068	3	0.49	this study	1.19	F	0.41
Cebus sp. https://doi.org/10.17602/M2/M62066	3	0.38	this study	0.85	F	0.45
Tupaia sp. https://doi.org/10.17602/M2/M62067	2	0.17	this study	0.4	F	0.43
Average						0.51
Standard deviation						0.14

Table 1B
Measurements of vessel lumen and osteological foramen diameters for select species. Reliability refers to the method by which the lumen diameter was measured: 1 = from CT, x-ray, or synchrotron angiography images. When more than one study is cited, measurements are averages of values reported in each source. Ratio = [lumen radius/foramen radius]. Sources: B—Findlay et al. (1988), C—Macdonald et al. (1991), E—Jeppsson and Olin (1960), F—Boyer and Harrington (2018), G—Schöning et al. (1994), H—Kane et al. (1996), I—Krejza et al. (2006), J—Scheel et al. (2000), K—Kidoguchi et al. (2006), L—Schambach et al. (2009), M—Cai et al. (2012).

Species	Internal carotid artery			Carotid canal/foramen		Ratio
	Reliability	Lumen radius (mm)	Sources	Radius (mm)	Sources	
Homo sapiens	1	2.35	G, H, I, J	3.34	F	0.70
Macaca fascicularis	1	0.77	B, C	1.05	F	0.73
Mus musculus	1	0.09	K, L	0.14	F, this study	0.64
Rattus norvegicus	1	0.30	M	0.36	F	0.83
Oryctolagus cuniculus	1	0.34	E	0.39	F, this study	0.87
Average						0.76
Standard deviation						0.09

concern to some degree. On the other hand, having large vena commitantes in the transverse foramen does not seem to lead to a greater offset between canal size and arterial vessel size as Seymour and Snelling (2018) suggest. We say this because *Macaca* does not have exceptionally small vertebral arteries for its transverse foramen size (Table 1A), even though the group to which it belongs (Old World monkeys) has been noted for having large vena commitantes in its transverse foramen (Seymour and Snelling, 2018).

Next, we computed a predictive equation for vessel wall shear stress (τ) of the vertebral artery following the approach used by Seymour and colleagues (2015, 2016, 2017) for the ICA. Using vertebral artery blood flow rates from Symon et al. (1963) and Table S1 of Boyer and Harrington (2018), vertebral artery lumen radius from Table 1A, and body masses from Seymour et al. (2015) for the human, monkey, and rat, we derived the scaling relationship: $\tau = 285\ M_b^{-0.22}$ for predicting wall shear stress in the vertebral arteries (τ_{VA}). Using τ_{VA} and the equation $Q = \frac{\tau\pi r^3}{4\eta}$, where η = blood viscosity (assumed to be a constant 0.04 dynes*s/cm) and r = vessel lumen radius (estimated as transverse foramen radius * 0.51), we predicted blood flow rate (Q) through the vertebral arteries. ICA blood flow rate was calculated using carotid canal radii from Boyer and Harrington (2018) and the methods of Seymour et al. (2015). Values for the variables mentioned above are provided for every species in the study of Boyer and Harrington (2018) in Supplementary Online Material (SOM) Table S6.

Given the critique by Seymour and Snelling (2018), there were four questions that we wanted to test using flow rates instead of arterial canal areas. First, does the vertebral artery flow rate help explain significant amounts of variation in forebrain size among euarchontans or is ICA flow sufficient? Second, after summing flow rates from the ICA and vertebral artery (total blood flow rate), do haplorhines and strepsirrhines have different scaling relationships between encephalic flow and brain size? Third, does total blood flow rate explain significant amounts of variance in measurements of brain metabolism or are brain size and/or neuron counts sufficient? Lastly, when predicting relative whole brain metabolic demands from total blood flow rate and brain size, are humans still within the range of other euarchontans, or do they appear more unusual?

4. Results and discussion

4.1. Vertebral artery flow is significantly correlated with forebrain mass

Re-running phylogenetic multiple regression with forebrain mass as the dependent variable and ICA and vertebral artery flow rates as independent variables confirms that both arteries explain significant amounts of variance in forebrain mass (SOM Table S1). In addition, we re-assessed whether predicted flow rates of the vertebral artery help explain forebrain size when controlling for body size (instead of ICA flow). The results of this analysis (SOM Table S2) show that vertebral artery flow has a significant correlation with brain part size, while the influence of body size is reduced compared to findings of Boyer and Harrington (2018). These results suggest that the vertebral artery is important for supplying the forebrain with blood. Conclusions from Boyer and Harrington (2018) are therefore upheld and arguably strengthened due to a greater number of significant post-hoc, subgroup comparisons and by the weakened effect of body mass.

4.2. Total blood flow rate scales similarly with brain size in haplorhines and strepsirrhines

Phylogenetic ANCOVA comparing non-cheirogaleid lemuriforms (where all flow can be considered as coming from the vertebral artery – Boyer et al., 2016) and haplorhines, shows no difference between intercepts (p = 0.99, t = 0.007) or slopes (p = 0.64, t = 0.10). Thus, Boyer and Harrington's (2018) conclusion that differences in cognitive complexity have not affected blood flow requirements or scaling in these clades is upheld. This result can be appreciated graphically from the similarly tight correlations of total canal area (Fig. 1A) and total flow rate (Fig. 1B) to brain size: if using canal area had been misleading here, flow rates should have shown more offset between strepsirrhines and haplorhines.

It is notable that the scaling exponent of total blood flow rate to brain size for the combined sample of haplorhine and non-cheirogaleid lemuriform primates is 0.86 (p < 0.0001, t = 26.90, 95% confidence interval = 0.81–0.92), because it matches the exponent of 0.86 derived for the scaling of empirically measured brain metabolic rate to brain volume (Karbowski, 2007). The confidence interval also overlaps the exponents derived for scaling of regional blood flow to brain structure volume (about 0.84) reported by Karbowski (2011).

In our haplorhine primate sample, ICA flow rate scales to ECV to the power of 0.98 (Fig. 2), similar to the results of Seymour and colleagues (2015, 2018), while our estimate of total flow rate (summing ICA and vertebral artery flow) still scales at 0.87 (95% confidence interval = 0.80–0.93) to brain size. For completeness, we also looked at total blood flow scaling to brain size in Euarchonta (the group including primates, treeshrews and dermopterans) and found a slope of 0.88 (95% confidence interval = 0.83–0.93) (Fig. 1B).

4.3. Predicted total encephalic blood flow rate correlates strongly with measured whole brain metabolism

For a sample of seven species (SOM Table S3; data from Boyer and Harrington, 2018: their table 2), we computed total blood flow rate and used multiple regression to assess whether blood flow rate accounts for significant variance in brain metabolism when controlling for endocranial volume and neuron counts. We found that total brain blood flow rate has the highest t-value and lowest p-value in multiple regression, meaning it explains more variance in brain metabolism than either endocranial volume or neuron count (SOM Table S4). Neuron counts are not significant in this model (p = 0.31), supporting the conclusion of Boyer and Harrington (2018) that data on neuron counts may not be critical for predicting brain metabolism.

We wish to reiterate that in bivariate regressions, neuron counts continue to correlate strongly with brain metabolism, as first shown by (Herculano-Houzel, 2011). What our results indicate is that variance in neuron count does not help significantly to explain variance in brain metabolism when controlling for endocranial volume and blood flow rate. We do not take our results to mean that neurons are causally unrelated to brain energy demands or that, with larger or different samples, neuron counts may ultimately prove more strongly correlated with brain metabolism than blood flow rate and endocranial volume. Our point is that neuron counts are not the strongest correlates or best predictors of brain metabolism with the currently available data. Even if neuron counts eventually edge out blood flow rate estimates and endocranial volume as predictors of brain metabolism, we think the osteological variables will remain important for predicting brain

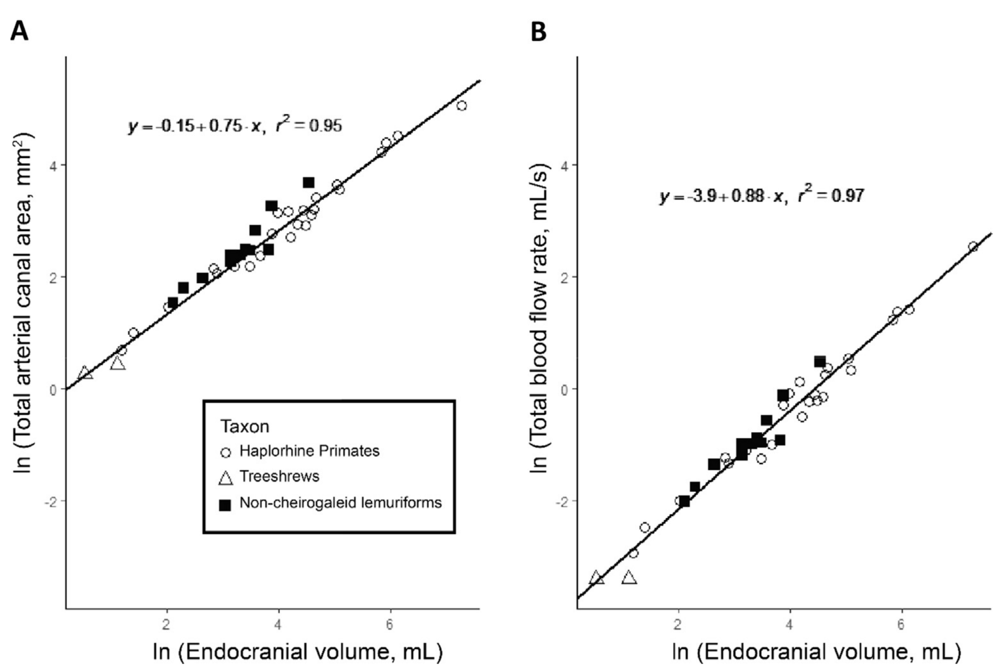

Figure 1.. Effect of replacing canal areas with flow rates in plots of Boyer and Harrington (2018). Phylogenetic generalized least squares regression analysis of euarchontans shows that total blood flow rate (mL/s) scales to endocranial volume (mL) with a slightly higher correlation coefficient than total arterial canal area (mm^2) due to reduced residuals for certain strepsirrhines. Total blood flow rate is the sum of flow rates predicted for the vertebral artery and internal carotid artery based on bony canal diameters for these vessels as described in the Methods section, while total arterial canal area is simply the sum of the area of the bony canals and is what Boyer and Harrington (2018) analyzed. The 95% confidence intervals on the slope of total blood flow rate to endocranial volume are 0.83–0.93.

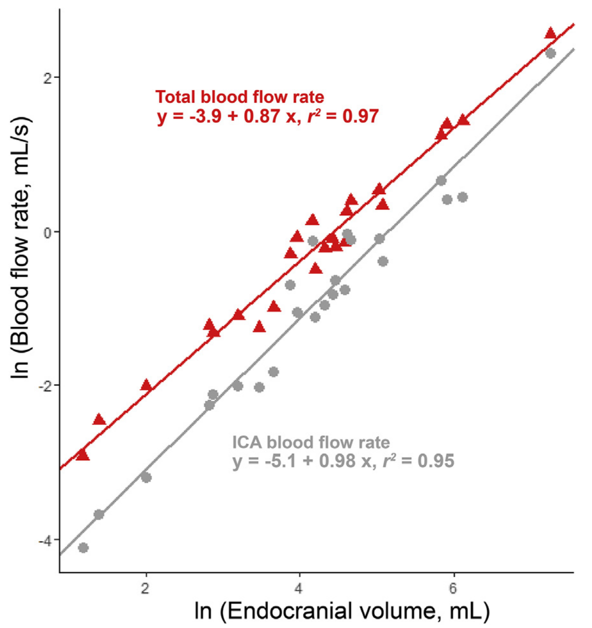

Figure 2. Role of the vertebral artery in total blood flow to the brain. In gray, we plot blood flow rates through the internal carotid artery (ICA) for a haplorhine sample as also done by Seymour and colleagues (2015, 2016, 2017, 2018). The slope of this line matches that recovered by Seymour's group and thereby reproduces their results. In red, we plot total blood flow rate for the same sample as estimated by summing ICA flow rate with vertebral artery blood flow rate. Vertebral artery flow rates take into account the finding that the vertebral artery does not fill the transverse foramen as completely as the internal carotid artery fills the carotid foramen (Table 1A, B). See Methods section for full details on calculating the vertebral artery flow rate. Note that total blood flow rate scales with negative allometry in haplorhines (the 95% confidence interval is 0.80–0.93), with an exponent and confidence interval similar to that recovered for the full euarchontan sample (Fig. 1) and similar to the exponent recovered by regressing brain metabolic energy consumption against brain size directly (Karbowski, 2007). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

metabolism in extinct and extant species for which soft tissue is unavailable.

4.4. When predicted from brain size and blood flow rate, human relative brain cost is not exceptional

Boyer and Harrington (2018) used the Bayesian phylogenetic prediction method of Nunn and Zhu (2014) to predict brain metabolic costs from measured encephalic arterial canal areas. We revisited the question of brain costs using the approach of Nunn and Zhu (2014), replacing measured canal area with predicted blood flow rate as a predictor variable. For the new approach using flow rates in place of canal area, we saw little change in levels of prediction error, which remained very low (4.5% jackknife), or 95% credibility intervals. The effect of the new approach on out-of-sample predictions of brain metabolism from Boyer and Harrington (2018) was minimal; average values of some predictions went up slightly while others went down (SOM Table S5). Importantly, non-cheirogaleid lemuriforms (which rely primarily on encephalic blood supply through the vertebral arteries) did not have vastly different mean predicted brain glucose utilization rates compared to the results of Boyer and Harrington (2018), which would be expected if use of the transverse foramen area (without accounting for the lesser degree of vessel filling) had been grossly overestimating the contribution of the vertebral artery to brain metabolism prediction. Gliroids are still predicted to have low relative brain costs compared to euarchontans. Certain treeshrews, lemurs, and New World monkeys were still predicted to have relative brain costs that match or exceed those in humans, and other catarrhines were still predicted to have relative brain costs substantially lower than those of humans (SOM Table S5). In sum, there were still no changes to the conclusions of Boyer and Harrington (2018).

We thank Seymour and Snelling (2018) for pointing out some issues with earlier studies by Boyer and colleagues (2016, 2018) and for giving us a venue for addressing them here. We find that support for the conclusions of Boyer and Harrington (2018) holds (and in many cases is strengthened) when using blood flow rate predictions instead of canal areas, even though the implied proportions of vertebral blood to ICA blood have decreased compared to their results.

We also feel this re-analysis has strengthened the idea that the rates at which brain blood flow and brain metabolic demands scale with brain size are not clade specific properties within mammals. Our results suggest values between 0.86 and 0.88 (95% CI: 0.80–0.93) for strepsirrhines, haplorhines, Primates, and Euarchonta. These slopes are extremely close to those recovered previously by Karbowski (2007) relating brain metabolism to brain size for a more diverse placental mammal sample. The hypothesis that blood flow rate scaling is uniform among mammals can be tested by regressing whole brain blood flow rate predictions on endocranial volume in marsupials, rodents and/or other clades where encephalic irrigation is completely accounted for by the internal carotid and vertebral arteries.

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Supplementary Online Material

Supplementary online material to this article can be found online at https://doi.org/10.1016/j.jhevol.2018.10.002.

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